

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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PDB Info	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	JOB ID: 3750189754
Assembled PDB:	step5_assembly.pdb (view structure)						download.tgz
Input Generator Input:	step5_input.inp						
Input Generator Output:	step5_input.out						
CHARMM Minimization:	step5_input_minimization.str						
Crystal Image:	crystal_image.str						
FFT Calculation:	checkfft.py						
Restraints:	membrane_restraint.str						Lipid Positional Restraint
	membrane_restraint2.str						Lipid Dihedral Restraint
Equilibration Inputs:	step6.1_equilibration.inp						Equilibration Step 1
	step6.2_equilibration.inp						Equilibration Step 2
	step6.3_equilibration.inp						Equilibration Step 3
	step6.4_equilibration.inp						Equilibration Step 4
	step6.5_equilibration.inp						Equilibration Step 5
	step6.6_equilibration.inp						Equilibration Step 6
Production Inputs:	step7_production.inp						Production Input

Equilibration Input Notes:

```
!
! Setup Restraints for Protein and Lipids (see @liptype_restraint.str)
!
! Suggested Equilibration Scheme [Reducing Force Constants]
! (5 Cycles, 1 cycle = 50 - 100 ps )
!-----
!      1 cycle    2 cycle    3 cycle    4 cycle    5 cycle    6 cycle
!-----
! BB      10.0      5.0       2.5       1.0       0.5       0.1
! SC       5.0      2.5       1.0       0.5       0.1       0.0
! wforce   2.5      2.5       1.0       0.5       0.1       0.0
! tforce   2.5      2.5       1.0       0.5       0.1       0.0
! mforce   2.5      2.5       1.0       0.5       0.1       0.0
! ion     10.0      0.0       0.0       0.0       0.0       0.0
!-----
!
! Equilibration
!-----
! To reduce the possible problem with the numerical integration with
! the uncorrelated system, 1 fs time-step is used only for the first-step of
! equilibration.
```

! It is still possible that you may need to use 1 fs for the next equilibration
! steps if your system is initially very very unstable (rare cases).
!
! ** Note: change "nstep" to reduce the number of dynamics steps



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